COMPUTER SCIENCE – FORM ONE

CHAPTER FIVE: COMPUTER SYSTEM HANDLING AND CARE

Teacher's Complete Simple Teaching Summary — Pages 102–113

Purpose: This summary is designed for classroom presentation. Each concept has a simple explanation, an easy real-life example and an important point for students to remember.

1. MEANING OF COMPUTER HANDLING AND CARE

Computer handling and care means taking care of a computer system and its accessories so that they can work properly, reliably and efficiently.

Teacher explanation: Computer care means using, cleaning, protecting and maintaining a computer properly so that it does not get damaged and continues to work well.

Example: A laptop can be damaged if a student pours water on it, hits the keyboard, uses it in excessive heat or installs harmful software.

Remember: If we take care of a computer, it can serve us for longer.

2. LIQUID DAMAGE

Liquid can seriously damage computer hardware. Liquid can interfere with electronic circuits and may cause a short circuit.

Teacher explanation: Computers use electricity, while water can conduct electricity. If liquid reaches electronic components, electricity may flow through an unintended path and damage the components.

Example: Water enters a laptop through the keyboard and the laptop stops working.

Remember: Keep water, juice, tea and other liquids away from computers.

3. CORROSION AND DATA LOSS

Liquid can also cause corrosion of metal parts. It can damage storage devices such as hard drives and may cause loss of access to stored data.

Teacher explanation: Corrosion is the gradual damage of metal caused by moisture or other substances. If a storage device is damaged, important files may become inaccessible.

Example: A student stores an assignment only on a laptop. Liquid damages the storage device and the assignment cannot be opened.

Remember: Hardware damage can lead to data loss.

4. WHAT TO DO AFTER A LIQUID SPILL

Turn off the computer, disconnect it from electricity and disconnect external devices. For a small external spill, carefully dry the accessible surface with a soft absorbent cloth. For serious spills, seek qualified technical help.

Teacher explanation: Never continue using a wet computer. Safety comes first.

Easy memory: OFF → UNPLUG → DISCONNECT → DRY CAREFULLY / GET HELP.

5. PREVENTING LIQUID DAMAGE

Keep drinks away from computers and use a clean, dry and safe working environment.

Example: Do not put a cup of tea next to a laptop or carry an open bottle over a computer.

Remember: Prevention is better than repairing liquid damage.

6. POWER MANAGEMENT

Computers require a suitable and stable supply of electricity. Power problems can cause hardware damage, malfunction and data loss.

Teacher explanation: Power management means taking measures to protect a computer from unstable electricity.

Main power problems: Brownouts, spikes and surges.

7. BROWNOUT

A brownout is a reduction in electrical voltage that lasts for a period. The book states that a brownout occurs when the voltage drops below 80% of the normal voltage level and gives Tanzania's normal voltage as 220V–240V.

Teacher explanation: A brownout means electricity becomes weaker than normal for some time.

Example: Lights become dim and a fan may rotate more slowly.

Remember: BROWNOUT = LOW VOLTAGE.

8. SPIKE

A spike is a sudden increase in voltage that lasts for a very short time. The book relates spikes to events such as lightning and power returning after a blackout.

Teacher explanation: A spike is a quick sudden increase in electrical voltage.

Example: Electricity goes off and then returns suddenly, causing a brief voltage increase.

Remember: SPIKE = SUDDEN INCREASE.

9. POWER SURGE

A power surge is a sudden and significant increase in electrical voltage that can damage electronic equipment.

Teacher explanation: A surge is a large increase in voltage above the normal level.

Example: An electrical disturbance may send excessive voltage to a computer and damage its power supply or other components.

Remember: SURGE = LARGE SUDDEN VOLTAGE INCREASE.

10. UPS – UNINTERRUPTIBLE POWER SUPPLY

A UPS provides temporary backup power when the main electricity supply fails.

Teacher explanation: A UPS is like a temporary battery for a computer. It gives the user time to save work and shut down safely.

Example: Electricity goes off while a student is typing an assignment. The UPS keeps the computer running temporarily so the student can save the work.

Remember: UPS = BACKUP POWER.

11. CONFIGURING AND USING A UPS

Check that the UPS can support the computer's power requirements, place it in a suitable ventilated location, connect it according to the manufacturer's instructions, connect the computer to the appropriate outlet and test it safely.

Teacher explanation: A UPS must be correctly sized and connected. Do not overload it.

Example: Connect the computer and monitor to suitable UPS outlets and follow the manufacturer's instructions for testing.

12. SURGE PROTECTOR

A surge protector is designed to help protect electrical equipment from voltage spikes and surges.

Teacher explanation: It helps prevent excessive voltage from reaching sensitive computer equipment.

Example: A computer connected through a suitable surge protector has an additional layer of protection against voltage surges.

Remember: SURGE PROTECTOR = PROTECTS FROM SURGES.

13. VOLTAGE STABILISER

A voltage stabiliser helps maintain a more stable voltage supplied to equipment when the incoming voltage changes.

Teacher explanation: It helps keep the voltage supplied to the computer more stable.

Example: If the incoming electricity fluctuates, a suitable stabiliser can help provide a steadier output.

Remember: STABILISER = STABLE VOLTAGE.

14. DIFFERENCE BETWEEN UPS, SURGE PROTECTOR AND STABILISER

UPS provides temporary backup power. A surge protector helps protect against voltage spikes and surges. A voltage stabiliser helps maintain stable voltage.

Easy classroom statement: UPS gives backup power; surge protector protects from surges; stabiliser helps keep voltage stable.

15. MOVING OR RELOCATING A COMPUTER

Before moving a computer, shut it down properly and disconnect its cables. Use suitable packaging or a carrying case and protect the equipment from physical damage.

Teacher explanation: Do not move a computer carelessly while it is operating.

Example: Disconnect the monitor, keyboard, mouse, power and network cables before moving a desktop computer.

16. LAPTOP HANDLING

Do not carry a laptop by its screen. Hold it securely from the base. Keep heavy, sharp or liquid objects away from it during transportation.

Example: Carry the laptop with both hands under the base instead of holding the display.

17. COMPUTER SOFTWARE CARE

Computer care also includes protecting software and data from harmful software and software problems.

Main areas: Viruses, antivirus protection and software updates.

18. COMPUTER VIRUSES

A computer virus is harmful software that can enter a computer and cause problems. Harmful software can spread through infected files, removable storage and unsafe downloads.

Teacher explanation: A virus can interfere with normal computer operation and may affect files and data.

Example: An infected USB drive is connected to a school computer and harmful software spreads to the computer.

19. ANTIVIRUS

Antivirus software helps detect and protect computers from viruses and other malicious software.

Good practice: Scan USB drives, scan downloaded files, scan the computer regularly and keep antivirus software updated.

Example: Connect a USB → scan it → check the result → open files only when safe.

20. SOFTWARE UPDATES

Operating systems, drivers and applications should be updated. Updates may provide security patches, bug fixes and improved operation.

Teacher explanation: An update can repair software problems and fix security weaknesses.

Example: Install genuine security updates when the operating system provides an important update.

21. PROPER COMPUTER SHUTDOWN

Proper shutdown means saving work, closing applications and using the operating system's shutdown procedure before powering off.

Easy memory: SAVE → CLOSE → LOG OUT → SHUT DOWN → WAIT.

Why? Proper shutdown helps reduce data loss, file corruption and system problems.

22. HANDLING COMPUTER PERIPHERALS

Peripheral devices include equipment such as keyboards, mice, printers, scanners and other devices connected to a computer. They should be handled carefully.

Keyboard: Type gently and do not put heavy objects on it.

Mouse: Handle it gently, use a suitable surface and do not drop or throw it.

23. CABLE CARE

Do not excessively bend cables, pull cables by the wire, place them near sharp objects or put heavy objects on them. Organise cables and inspect them for damage.

Correct method: Hold the connector when disconnecting a cable.

Safety: Never use a cable with exposed electrical wires.

24. PASSWORD PROTECTION

A password is a secret combination used to control access to a computer or account.

Teacher explanation: A password is like a key. Only authorised users should know it.

Example: A school computer account may require a password before a student can access it.

25. STRONG PASSWORDS

Use a combination of uppercase letters, lowercase letters, numbers and special characters. Longer passwords are generally harder to guess.

Example for teaching only: School@2026#Lab. Students should create their own private passwords.

Avoid: Your name, birthday, 123456, password or other obvious choices.

26. UNIQUE PASSWORDS

Do not use exactly the same password for every important account.

Example: If the same password is used for email and another account and it becomes known, more than one account may be at risk.

27. PASSWORD MANAGERS

A password manager can help store passwords securely and generate strong passwords.

Teacher explanation: It is like a secure digital place for managing many passwords.

28. TWO-FACTOR AUTHENTICATION (2FA)

2FA adds another verification step after the password, such as a verification code.

Example: Enter password → receive verification code → enter code → access account.

Memory: Password = first lock; 2FA = second lock.

29. PHISHING AWARENESS

Do not enter passwords into suspicious websites or give them to people through suspicious messages. Verify unexpected requests before responding.

Example: A message says, 'Your account will close today; enter your password here.' Do not immediately trust the link.

30. DATA BACKUP

A backup is an additional copy of important data. Backups can help recover data after hardware failure, accidental deletion or other data-loss events.

Example: Keep another copy of an important school assignment so that it can be recovered if the computer fails.

Remember: BACKUP = EXTRA COPY.

31. PHYSICAL SECURITY

Physical security means protecting computer equipment from theft, unauthorised physical access and physical damage.

Example: Keep a school computer laboratory locked when it is not in use and control who can enter.

32. CABLE LOCKS

A cable lock can attach a computer to a fixed object, making it more difficult to remove or steal.

Example: A laptop in an office can be secured to a desk using a security cable.

33. SECURE COMPUTER PREMISES

Physical security can include locked doors, CCTV, alarms and access control systems.

Teacher explanation: Computer security is not only about passwords. The physical place where computers are kept must also be protected.

34. COMPLETE CONCEPT MAP

COMPUTER CARE

→ **Hardware care:** protect from liquid, dust, heat and physical damage.
→ **Power protection:** understand brownouts, spikes and surges; use UPS, surge protection and voltage stabilisation where appropriate.
→ **Software care:** antivirus and software updates.
→ **Safe use:** proper shutdown and careful handling of peripherals and cables.
→ **Account security:** strong unique passwords and 2FA.
→ **Data protection:** regular backups.
→ **Physical security:** secure rooms, access control, CCTV, alarms and cable locks.

35. TEN MOST IMPORTANT POINTS FOR STUDENTS

1. Keep liquids away from computers.

2. Protect computers from unstable electricity.

3. UPS provides temporary backup power.

4. A surge protector helps protect against spikes and surges.

5. A voltage stabiliser helps maintain stable voltage.

6. Use antivirus and keep software updated.

7. Shut down the computer properly.

8. Handle keyboards, mice and cables carefully.

9. Use strong, unique passwords and 2FA where possible.

10. Back up important data and protect computers physically.

36. QUICK CLASSROOM QUESTIONS

1. What is computer handling and care?

2. Why is liquid dangerous to a computer?

3. What is a brownout?

4. What is a power spike?

5. What is a power surge?

6. What is the purpose of a UPS?

7. What is the difference between a UPS and a voltage stabiliser?

8. What does a surge protector do?

9. Why should software be updated?

10. What can happen if a computer is shut down forcefully?

11. How should a keyboard and mouse be handled?

12. How should computer cables be handled?

13. What makes a password strong?

14. What is 2FA?

15. What is a backup?

16. What is physical security?

17. Give three ways of protecting a computer laboratory.

37. ONE-LINE REVISION ANSWERS

Brownout	A temporary reduction in electrical voltage.
Spike	A sudden short increase in voltage.
Surge	A sudden significant increase in voltage.
UPS	A device that provides temporary backup power when the main power fails.
Surge protector	A device that helps protect equipment from voltage spikes and surges.
Voltage stabiliser	A device that helps maintain a more stable voltage supplied to equipment.
Antivirus	Software used to detect and protect against malicious software.
Backup	An additional copy of important data.
2FA	A security method requiring a second verification step in addition to a password.
Physical security	Measures used to protect computer equipment from theft, unauthorised physical access and physical damage.
Proper shutdown	Saving work, closing programs and using the operating system's shutdown procedure before powering off.

FINAL TEACHER STATEMENT

"Computer handling and care means taking proper care of the computer, its accessories, software and information. We protect computers from liquids, unstable electricity, viruses, physical damage and theft. We use suitable power protection, keep software updated, shut down computers properly, handle peripherals and cables carefully, use strong passwords and two-factor authentication, back up important data and keep computers in secure places."

Simple final message: Protect the computer, protect the software, protect the data, and protect the user.